$(x+y)\,\text{kJmol}^{-1}$

$(x+y)\times 10^{3}\ \text{Jmol}^{-1}$

11. **வாய்ப்பு(இ)**

$T_1 = 200K\ ;\ k = k_1$

$T_2 = 400K\ ;\ k = k_2 = 2k_1$

$$\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1\ T_2}\right)$$

$$\log\left(\frac{2\,k_1}{k_1}\right) = \frac{E_a}{2.303\times 8.314\ \text{J K}^{-1}\text{mol}^{-1}}\left(\frac{400\ \text{K} - 200\text{K}}{200\text{K}\ \times\ 400\text{K}}\right)$$

$$E_a = \frac{0.3010\times\ 2.303\ \times\ 8.314\ \text{JK}^{-1}\,\text{mol}^{-1}\times\ 200\ \text{K}\times 400\ \text{K}}{200\ \text{K}}$$

$E_a = 2305\ \text{J mol}^{-1}$

$E_a = 2.305\ \text{kJ mol}^{-1}$

12. **வாய்ப்பு(அ)**

$$k = \left(\frac{2.303}{t}\right)\log\left(\frac{[A_0]}{[A]}\right)$$

$$2.303\ \times\ 10^{-2}\ \text{hour}^{-1} = \left(\frac{2.303}{1806\ \text{min}}\right)\log\left(\frac{0.25}{[A]}\right)$$

$$\left(\frac{2.303\ \times\ 10^{-2}\ \text{hour}^{-1}\times 1806\ \text{min}}{2.303}\right) = \log\left(\frac{0.25}{[A]}\right)$$

$$\left(\frac{1806\ \times 10^{-2}}{60}\right) = \log\left(\frac{0.25}{[A]}\right)$$

$$0.301 = \log\left(\frac{0.25}{[A]}\right)$$

$$\log 2 = \log\left(\frac{0.25}{[A]}\right)$$

$$2 = \left(\frac{0.25}{[A]}\right)$$

$$[A] = \left(\frac{0.25}{2}\right) = 0.125\text{M}$$

13. **வாய்ப்பு(ஆ)**

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